



AWS Academy Cloud Foundations – Configuring Identity and Access Management, Virtual Private Cloud, and Elastic Compute Cloud

Yuuki Burleigh, CCNA

Lab 1 – Introduction to AWS IAM

Purpose

This lab helps to develop an understanding of how to secure the cloud using Identity based access policies. Basic access security concepts covered in the lab contribute to developing an understanding of how to secure access in networks beyond the cloud and into traditional networks and inside applications. The cloud security concepts covered in the lab also transfer to how to secure other resources within the cloud, like the best practice of least privilege a practice used to limit access to the minimum required resources for the action or role.

Background

Amazon IAM or Identity Access Management is a way to secure access to resources within AWS. IAM uses username and password login credentials to prove a client's identity and give the specified access that IAM specifies for the provided login credentials. For example, if someone were hired to optimize S3 lifecycle policies within an organization, best practice for the organization would be to provide login credentials that IAM specifies can only alter S3 lifecycle policies and nothing else. This practice of creating IAM users that only have permission to access or edit the specific resources they need within the cloud is called the principle of least privilege.

The principle of least privilege is an important concept beyond just IAM policies and best practices in the cloud. It is a basic access security concept to apply to traditional networks, applications, and even in everyday life. Providing the minimum amount of access required allow for companies and organizations that use third party services and segmentation of operations prevents extensive damage within the organization when an employee or hired third party tries to abuse their provided credentials, as the perpetrator is only given the access to resources they require. [An IBM report](#) states that 83% of organizations reported in 2024 that they had an insider attack, this further emphasizes the importance of practicing the principal of least privilege as the people who attack organizations and companies from the inside already have access to resources, and being able to limit their access to only what they were originally instead to do will prevent their ability to bring down the company or organization.

Using IAM, one AWS account can have multiple users under it, each with access to different resources. The level of specificity that IAM can achieve is very granular and fine-tuned, allowing organizations and companies to follow the principle of least privilege very closely. IAM uses IAM policies to achieve this effect. These policies apply to virtually all AWS services and resources; you can limit access to simply reading configurations or editing all services. These policies are written in a JSON (JavaScript Object Notation) format, AWS has JSON code for every resource within AWS which allows for the elevated level of granularity that allows for organizations to follow to principle of least privilege very closely. The JSON statements are formatted as permission statements, allowing access rather than denying things. If something that is not explicitly allowed is attempted on a user's account, it will be implicitly denied.

Another related service within AWS is AWS Organizations. AWS Organizations allow for large Organizations to create groups of users and limit access to resources using SCPs or Service Control Policies to explicitly deny that explicitly allow.

Lab 2 – Build Your VPC and Launch a Web Server

Purpose

This lab helps to develop an understanding of how to set up and connect resources within AWS using a Virtual Private Network or VPC through launching a web server in an EC2 Instance. This lab also contributes to developing an understanding of how logically isolated networks and containerized applications function and communicate with other resources in real time.

Background

A VPC or Virtual Private Cloud is a logically isolated network hosted in AWS that sits inside a singular AWS region but can be spread across multiple availability zones or AZs. Each VPC comes with 2 subnets, a public subnet, and a private subnet. Each subnet is allocated to an IP address range which any resources within it will be assigned one of the IP addresses within the range. The public subnet has a public IP address range so resources outside the VPC can be accessed by resources inside it, access from the outside is also allowed. Resources from the private subnet can be accessed if within the same VPC but cannot be accessed from outside the VPC without being configured, if a NAT gateway and route to the gateway is configured for the private subnet in the VPC the resources inside the subnet can initiate outbound connection but cannot be accessed from the outside. In most use cases, the private subnet is used for processing requests from the public subnet who gets requests from users.

Along with subnets, when creating a VPC it automatically comes with an Internet Gateway, Route Table, Security Group, and a Network ACL. The internet gateway is automatically installed within the VPC to provide internet access to the resources within the public subnet and does not need a NAT gateway like the private subnet as all resources in the public subnet are assigned a public IP address. Along with the internet gateway, the main route table is installed in the VPC, which comes with a default route that points all traffic to the Internet Gateway if the destination is not within the VPC. The Route table comes with a default Local Route that handles all traffic within the VPC subnets. A default security group that applies to the VPC is also created which allows for access control on what can interact with the VPC. The security group can be used in tangent with IAM policies to allow or deny access to editing the VPC using IAM. A network ACL is like a security policy or IAM policy but only applies to the VPC network as ACLs use IP addresses for Access Control.

Lab 3 – Introduction to Amazon EC2

Purpose

This lab builds the understanding of what EC2 instances are and how they are fundamental to providing any usable service or solution in the AWS cloud. EC2 instances are the basic units of computing in the cloud. This lab helps develop that

understanding and serves as an introductory step to understanding autoscaling and elastic load balancing.

Background

An EC2 instance is a basic cloud computing unit to process information for cloud solutions. EC2 instances can vary in size and type, with each type/size varying in cost. Depending on the use case, the type of EC2 instance to choose for the cloud solution will vary. T/M EC2 type is good for general purpose tasks, Type C are compute optimized and should be used for high-performance processes which, or solutions that need to deliver results fast. Type I/D are storage optimized and should be used for solutions that require fast performance and large datasets.

EC2 instances have lots of flexibility in purchasing options, On-Demand Instances, Spot Instances, and Reserved Instances. On-demand instances are the most basic and are paid by the second and require no minimum lease or any level of commitment. Spot Instances tend to be the most cost effective as they utilize unused EC2 capacity that other people are not using, although only 5% of spot instances are terminated monthly, these instances can be cut off by the providers only with a 2-minute warning or be cut off if you were outbid for the spot instance. With this drawback spot instances can save up to 90% in costs. Reserved instances are reliable but require long term commitment for typically 1-3 years but typically can save up to 72% in costs. AWS also allows for Dedicated Instances and Dedicated Hosts as well. A Dedicated Instance runs hardware dedicated to a single customer and provides isolation if needed for compliance and regulatory needs. A Dedicated Host takes that step further and dedicates a full physical server to a single customer, offering complete control over where you place your instance in the server and is beneficial for compliance and scenarios where you need your own license.

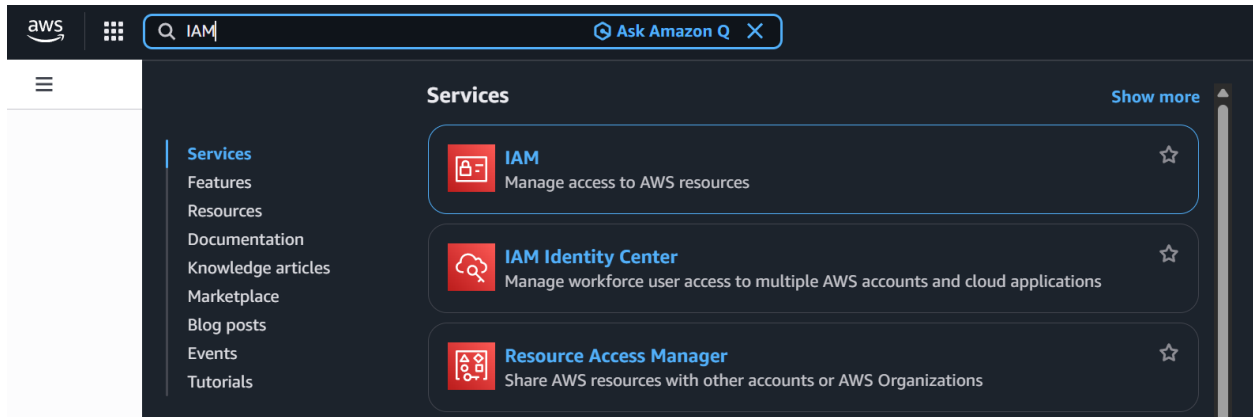
Lab 1 – Introduction to AWS IAM - Walkthrough

Navigate to Modules in AWS Academy and finish all the instructional content leading up to the first lab

Then, click on the lab and press start lab: 

Then, click “AWS” once the icon turns green: 

You are now in the AWS console. Then, click the search box in the top left of the AWS console and search IAM and click on it.



In the left navigation pane, choose Users, notice how there are 3 Users already listed.

Users (4) [Info](#)

An IAM user is an identity with long-term credentials that is used to interact with AWS in an account.

☐	User name	Path	Group:	Last activity	MFA	Password age
☐	awsstudent	/	Access denied	-	Access denied	Access denied
☐	user-1	/spl66/	0	-	-	✓ 4 minutes
☐	user-2	/spl66/	0	-	-	✓ 4 minutes
☐	user-3	/spl66/	0	-	-	✓ 4 minutes

Choose the User-1 Link, notice how there are no policies attached to this user

user-1 [Info](#) [Delete](#)

Summary

ARN
[arn:aws:iam::758503910673:user/spl66/user-1](#)

Console access
[Enabled without MFA](#)

Access key 1
AKIA3BG53FUI3FA6OVWK - Active
[Never used. Created today.](#)

Created
February 02, 2026, 13:25 (UTC-08:00)

Last console sign-in
[Never](#)

Access key 2
[Create access key](#)

[Permissions](#) [Groups](#) [Tags \(1\)](#) [Security credentials](#) [Last Accessed](#)

Permissions policies (0) [Refresh](#) [Remove](#) [Add permissions](#)

Permissions are defined by policies attached to the user directly or through groups.

Filter by Type
All types

☐ Policy name

☐ Type

☐ Attach...

No resources to display

Permissions boundary (not set)**Generate policy based on CloudTrail events**

You can generate a new policy based on the access activity for this user, then customize, create, and attach it to this role. AWS uses your CloudTrail events to identify the services and actions used and generate a policy. [Learn more](#)

[Generate policy](#)

No requests to generate a policy in the past 7 days.

Now navigate to Groups on User-1, Notice how User-1 is not a part of any group as well.

user-1 [Info](#) [Delete](#)

Summary

ARN
[arn:aws:iam::758503910673:user/spl66/user-1](#)

Console access
[Enabled without MFA](#)

Access key 1
AKIA3BG53FUI3FA6OVWK - Active
[Never used. Created today.](#)

Created
February 02, 2026, 13:25 (UTC-08:00)

Last console sign-in
[Never](#)

Access key 2
[Create access key](#)

[Permissions](#) [Groups](#) [Tags \(1\)](#) [Security credentials](#) [Last Accessed](#)

User groups membership [Remove](#) [Add user to groups](#)

A user group is a collection of IAM users. Use groups to specify permissions for a collection of users. A user can be a member of up to 10 groups at a time.

☐ Group name

☐ Attached policies

No resources
This user does not belong to any groups.

Now Navigate to the Security Credentials Tab and notice how User-1 is assigned a console password

user-1 [Info](#) [Delete](#)

Summary

ARN
[arn:aws:iam::758503910673:user/spl66/user-1](#)

Console access
[Enabled without MFA](#)

Access key 1
AKIA3BG53FUI3FA6OVWK - Active
[Never used. Created today.](#)

Created
February 02, 2026, 13:25 (UTC-08:00)

Last console sign-in
[Never](#)

Access key 2
[Create access key](#)

[Permissions](#) [Groups](#) [Tags \(1\)](#) [Security credentials](#) [Last Accessed](#)

User groups membership [Remove](#) [Add user to groups](#)

A user group is a collection of IAM users. Use groups to specify permissions for a collection of users. A user can be a member of up to 10 groups at a time.

☐ Group name

☐ Attached policies

No resources
This user does not belong to any groups.

In the left navigation pane choose User Groups and notice the groups that are set up for you.

User groups (3) [Info](#)

A user group is a collection of IAM users. Use groups to specify permissions for a collection of users.

Q Search

☐	Group name	Users	Permissions	Creation time
☐	EC2-Admin		Defined	5 minutes ago
☐	EC2-Support		Defined	5 minutes ago
☐	S3-Support		Defined	5 minutes ago

Click each link and go to the permission tab to see what access is allowed for each IAM group

EC2-Support [Info](#) [Delete](#)

Summary [Edit](#)

User group name EC2-Support	Creation time February 02, 2026, 13:26 (UTC-08:00)	ARN arn:aws:iam::758503910673:group/spl66/EC2-Support
---------------------------------------	--	---

Users **Permissions** Access Advisor

Permissions policies (1) [Info](#) [Simulate](#) [Remove](#) [Add permissions](#)

You can attach up to 10 managed policies.

Q Search Filter by Type All types

☐	Policy name	Type	Attached entities
☐	AmazonEC2ReadOnlyAccess	AWS managed	1

Also notice that no users are in any groups in the users tab

To see how the JSON is structured go to the permissions tab and click the plus or link to the permissions policies

EC2-Support [Info](#) [Delete](#)

Summary [Edit](#)

User group name EC2-Support	Creation time February 02, 2026, 13:26 (UTC-08:00)	ARN arn:aws:iam::758503910673:group/spl66/EC2-Support
---------------------------------------	--	---

Users **Permissions** Access Advisor

Permissions policies (1) [Info](#) [Simulate](#) [Remove](#) [Add permissions](#)

You can attach up to 10 managed policies.

Q Search Filter by Type All types

☐	Policy name	Type	Attached entities
☐	AmazonEC2ReadOnlyAccess	AWS managed	1

AmazonEC2ReadOnlyAccess [Copy JSON](#)

```

1- {
2-   "Version": "2012-10-17",
3-   "Statement": [
4-     {
5-       "Effect": "Allow",
6-       "Action": [
7-         "ec2:Describe*",
8-         "ec2:GetSecurityGroupsForVpc"
9-       ],
10-      "Resource": "*"
11-    },
12-    {
13-      "Effect": "Allow",
14-      "Action": "elasticloadbalancing:Describe*",
15-      "Resource": "*"
16-    },
17-    {
18-      "Effect": "Allow",
19-      "Action": [
20-        "cloudwatch:ListMetrics",

```

Now add user-1 to the S3-Support group, go to the Users tab and click add users on the top right, and select User-1 to add to S3-Support.

Add users to S3-Support [info](#)

Other users in this account (1/4)

Q Search

< 1 > ⚙

☐	User name	Group	Last activity	Creation time
☐	awsstudent	Access denied	-	14 minutes ago
☒	user-1	0	-	15 minutes ago
☐	user-2	0	-	15 minutes ago
☐	user-3	0	-	15 minutes ago

Cancel

Add users

Now add User-2 to EC2-Support

Add users to EC2-Support [info](#)

Other users in this account (1/4)

Q Search

< 1 > ⚙

☐	User name	Group	Last activity	Creation time
☐	awsstudent	Access denied	-	16 minutes ago
☐	user-1	0	-	16 minutes ago
☒	user-2	0	-	16 minutes ago
☐	user-3	0	-	16 minutes ago

Cancel

Add users

Now add User-3 to EC2-Admin

Add users to EC2-Admin [info](#)

Other users in this account (1/4)

Q Search

< 1 > ⚙

☐	User name	Group	Last activity	Creation time
☐	awsstudent	Access denied	-	16 minutes ago
☐	user-1	0	-	17 minutes ago
☐	user-2	0	-	17 minutes ago
☒	user-3	0	-	17 minutes ago

Cancel

Add users

You can verify that there is one user in each group using the User Group tab on the left

User groups (3) [info](#)

A user group is a collection of IAM users. Use groups to specify permissions for a collection of users.

Q Search

< 1 > ⚙

☐	Group name	Users	Permissions	Creation time
☐	EC2-Admin	1	Defined	17 minutes ago
☐	EC2-Support	1	Defined	17 minutes ago
☐	S3-Support	1	Defined	17 minutes ago

You can also look in the IAM Dashboard

IAM Dashboard [Info](#)

IAM resources
Resources in this AWS Account

User groups	Users	Roles	Policies	Identity providers
3	4	20	1	0

What's new [Info](#)
Updates for features in IAM [View all](#)

- AWS IAM enables identity federation to external services using JSON Web Tokens (JWTs). 2 months ago
- Streamline integration with Amazon and AWS Partner products using AWS IAM temporary delegation. 2 months ago
- AWS IAM launches awsSourceProfile condition key for region-based access control. 2 months ago
- AWS Service Reference Information now supports SDK Operation to Action mapping. 2 months ago

[more](#)

AWS Account

Account ID
758503910673

Account Alias
[Create](#)

Sign-in URL for IAM users in this account
<https://758503910673.signin.aws.amazon.com/console>

Tools [Info](#)

Policy simulator
The simulator evaluates the policies that you choose and determines the effective permissions for each of the actions that you specify.

Additional information [Info](#)

[Security best practices in IAM](#)

[IAM documentation](#)

[Videos, blog posts, and additional resources](#)

Now Login to one of the Users using the link on the left of teh IAM Dashboard

Sign-in URL for IAM users in this account

 <https://758503910673.signin.aws.amazon.com/console>

Copy the URL and open the link in an incognito window or the e1uivelent of your browser.

Now sign-in as User-1 who is assigned to the S3-Support group

IAM username: user-1

Password: Lab-Password1

Once signed in, type S3 in the search bar and click the “S3” link

Search: [Ask Amazon Q](#) [X](#)

Services [Show more](#)

-  **S3**
Scalable Storage in the Cloud
-  **S3 Glacier**
Archive Storage in the Cloud
-  **AWS Snow Family**
Large Scale Data Transport

Services
[Features](#)
[Documentation](#)
[Knowledge articles](#)
[Marketplace](#)
[Blog posts](#)
[Events](#)
[Tutorials](#)

Click the Bucket and see if there is anything inside of it

samplebucket--aef6adf0 [Info](#)

[Objects](#) [Metadata](#) [Properties](#) [Permissions](#) [Metrics](#) [Management](#) [Access Points](#)

Objects (0) [Copy S3 URI](#) [Copy URL](#) [Download](#) [Open](#) [Delete](#) [Actions](#) [Create folder](#) [Upload](#)

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. [Learn more](#)

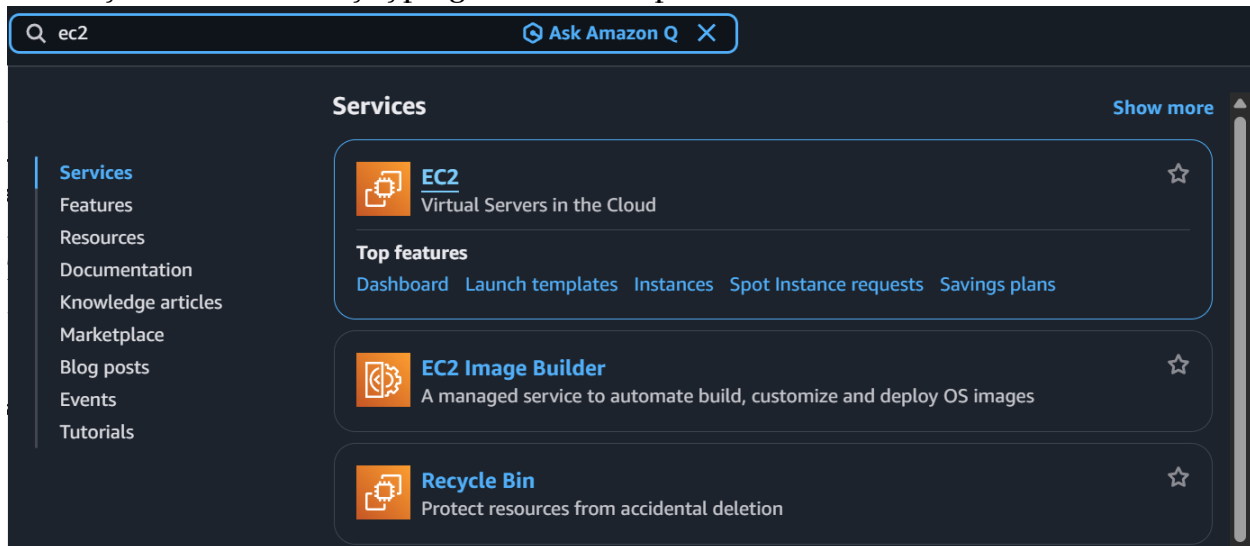
Name	Type	Last modified	Size	Storage class
No objects				

You don't have any objects in this bucket.

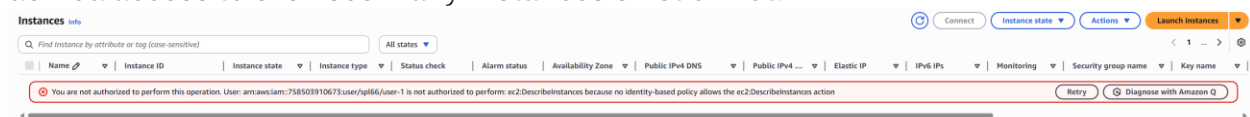
[Upload](#)

In this image there are no objects in samplebucket--aef6adf0

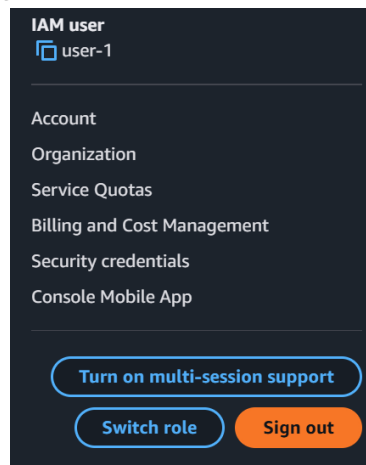
Now try to access EC2 by typing EC2 in the top left search bar



Now navigate to Instances using the left navigation pane and notice how you are denied access to even see if any instances exist or not.



Now sign user-1 out of the console by choosing user-1 on the top right of the screen and choosing “Sign Out”

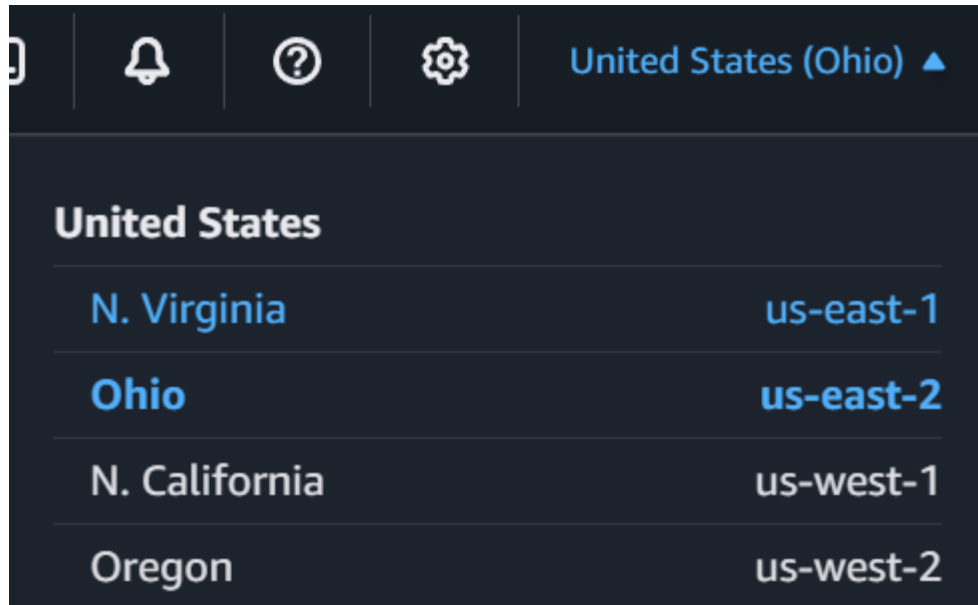


Now repeat the URL pasting process again in an incognito or equivalent window but now sign into user-2

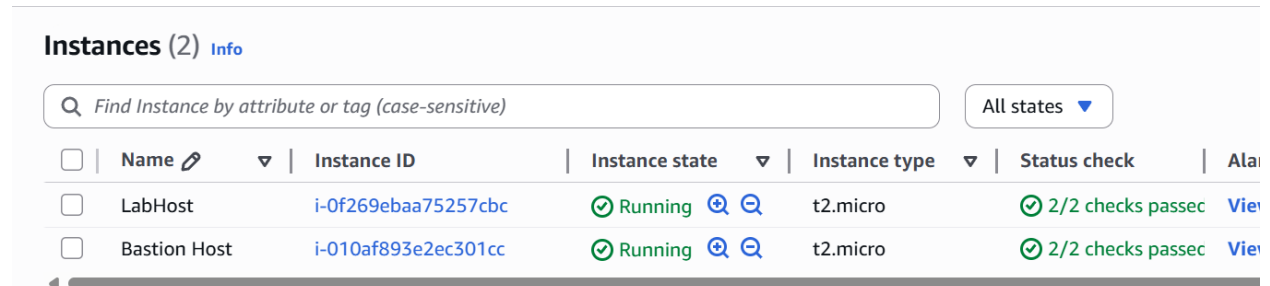
Username: user-2

Password: Lab-Password2

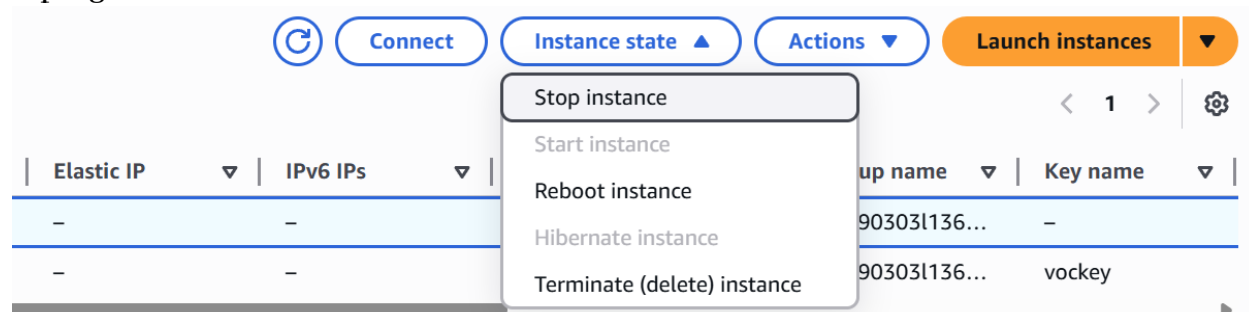
Search EC2 again and navigate to EC2 Instances and change your region in the top right to Virginia if it isn't already



Notice how user-2 can view all instances



Now select the lab host instance and stop it using the instance state menu in the top right



You should receive a failure message as User-2 is not authorized to edit anything but only view EC2 Instances, if you navigate to the S3 console you will see that you do not have permissions to view buckets as well.

Sign out and repeat the same process for stopping the EC2 instance for user-3

Username: user-3

Password: Lab-Password3

This time user-3 will succeed as they have permissions to stop EC2 instances

✓ Successfully initiated stopping of i-0f269ebaa75257cbc

Instances (1/2) [Info](#)

Find Instance by attribute or tag (case-sensitive)

All states ▼

☐	Name ✎	Instance ID	Instance state ▼	Instance type ▼	Status check	Al:
☒	LabHost	i-0f269ebaa75257cbc	⌚ Stopping 🔍 🔍	t2.micro	✓ 2/2 checks passed	✖
☐	Bastion Host	i-010af893e2ec301cc	✓ Running 🔍 🔍	t2.micro	✓ 2/2 checks passed	✖

Now go back to the Vocareuem Workpage submit the lab and get a full score, you can resubmit multiple times until you get a full score.

Total score

40/40

TASK 2a - Added user-1 to S3-Support group 5/5

TASK 2b - Added user-2 to EC2-Support group 5/5

TASK 2c - Added user-3 to EC2-Admin group 5/5

TASK 3a - user-1 logged in 5/5

TASK 3b - user-2 logged in 5/5

TASK 3c - user-2 ec2 stop instance attempt 5/5

TASK 3d - user-3 logged in 5/5

TASK 3e - user-3 EC2 stop instance attempt 5/5

Lab 2 – Build Your VPC and Launch a Web Server

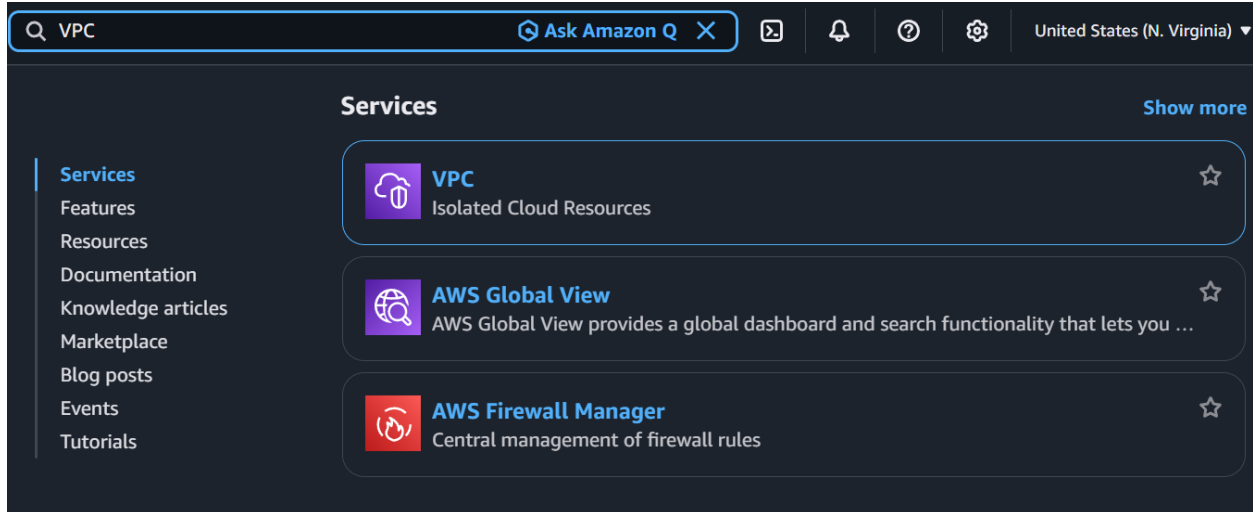
Repeat the steps to start the lab from Lab 1 and start the lab

In this lab you will create a VPC with 2 sets of public and private subnets with each set in a different availability zone with a NAT gateway in one public subnet and a web server in the other public subnet.

Now start by verifying that you are in the region N. Virginia (us-east-1)

United States (N. Virginia) ▼

Now type VPC in the console search bar and click VPC



Now Choose Create VPC and configure it with the following settings

Choose: VPC and More

Under Name tag auto-generation, keep auto-generate selection but change the name from project to lab

Keep the IPv4 CIDR Block to 10.0.0.0/16 these are going to be the IP addresses assigned to the resources in the subnets

For number of AZs or Availability zones to 1

For number of public and private subnets choose 1

Expand the Customize subnets CIDR blocks section

Change the public subnet CIDR block to 10.0.0.0/24

Change the private subnet CIDR block to 10.0.1.0/24

Set the NAT gateways to In 1 AZ

Set VPC endpoint to none

Keep both DNS hostnames and DNS resolution enabled

Create VPC [Info](#)

A VPC is an isolated portion of the AWS Cloud populated by AWS objects, such as

VPC settings

Resources to create [Info](#)

Create only the VPC resource or the VPC and other networking resources.

☐ VPC only ☒ VPC and more

Name tag auto-generation [Info](#)

Enter a value for the Name tag. This value will be used to auto-generate Name tags for all resources in the VPC.

☒ Auto-generate

lab

IPv4 CIDR block [Info](#)

Determine the starting IP and the size of your VPC using CIDR notation.

10.0.0.0/16 65,536 IPs

CIDR block size must be between /16 and /28.

IPv6 CIDR block [Info](#)

☒ No IPv6 CIDR block

Customize subnets CIDR blocks

Public subnet CIDR block in us-east-1a

10.0.0.0/24 256 IPs

Private subnet CIDR block in us-east-1a

10.0.1.0/24 256 IPs

NAT gateways (\$) - updated [Info](#)

NAT gateway allows private resources to access the internet from any availability zone within a VPC, providing a single managed internet exit point for the entire region. Additional charges apply.

☐ None ☐ Regional - new ☒ Zonal

NAT gateways (\$) [Info](#)

Choose the number of Availability Zones (AZs) in which to create NAT gateways.

In the right-side preview panel verify the configurations to the Lab VPC

VPC: lab-vpc

Subnets:

us-east-1a

Public subnet name: lab-subnet-public1-us-east-1a

Private subnet name: lab-subnet-private1-us-east-1a

Route tables

lab-rtb-public

lab-rtb-private1-us-east-1a

Network connections

lab-igw

lab-nat-public1-us-east-1a

Now Choose Create VPC and then View VPC

The screenshot shows the AWS VPC console interface. On the left is a navigation pane with categories: Virtual private cloud, Security, and PrivateLink and Lattice. The main content area is titled 'vpc-09fbc6db0db097363 / lab-vpc'. It features a 'Details' section with various attributes like VPC ID, State (Available), DNS resolution, Main network ACL, IPv6 CIDR, Encryption control ID, Tenancy, Default VPC, Network Address Usage metrics, Encryption control mode, Block Public Access, DHCP option set, IPv4 CIDR, Route 53 Resolver DNS Firewall rule groups, DNS hostnames, Main route table, and IPv6 pool. Below the details is a 'Resource map' section showing a diagram of the VPC, its subnets (lab-subnet-public1-us-east-1a and lab-subnet-private1-us-east-1a), and route tables (lab-rtb-public, lab-rtb-private1-us-east-1a, and rtb-0ea8583de8b960e7).

Now Create the second set of subnets; in the left navigation pane choose subnets

Choose Create Subnets and configure:

VPC ID: lab-vpc (select from the menu).

Subnet name: lab-subnet-public2

Availability Zone: Select the second Availability Zone (ex: us-east-1b)

IPv4 CIDR block: 10.0.2.0/24

The screenshot shows the 'Subnet settings' page in the AWS console. It includes fields for Subnet name (lab-subnet-public2), Availability Zone (us-east-1b), IPv4 VPC CIDR block (10.0.0.0/16), and IPv4 subnet CIDR block (10.0.2.0/24). There is also a section for Tags with a key 'Name' and value 'lab-subnet-public2'.

Create the Subnet; Repeat the process but change the name from “lab-subnet-public2” to “lab-subnet-private2” and change the IPv6 CIDR block to 10.0.3.0/24

Create subnet [Info](#)

VPC

VPC ID

Create subnets in this VPC.

vpc-09fbc6db0db097363 (lab-vpc)

Associated VPC CIDRs

IPv4 CIDRs
10.0.0.0/16

Subnet settings

Specify the CIDR blocks and Availability Zone for the subnet.

Subnet 1 of 1

Subnet name

Create a tag with a key of 'Name' and a value that you specify.

lab-subnet-private2

The name can be up to 256 characters long.

Availability Zone [Info](#)

Choose the zone in which your subnet will reside, or let Amazon choose one for you.

United States (N. Virginia) / use1-az1 (us-east-1b)

IPv4 VPC CIDR block [Info](#)

Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.

10.0.0.0/16

IPv4 subnet CIDR block

10.0.3.0/24

256 IPs

Now that the 2 subnets have been created, navigate to the Route Tables using the left navigation pane.

Select the route table for the first private subnet

Route tables (1/6) [Info](#)

Last updated
1 minute ago



Actions

Create route table

Find route tables by attribute or tag

< 1 > ⚙

	Name	Route table ID	Explicit subnet associations
☐	Work Public Route Table	rtb-0fd954ddf89d5a613	subnet-002ba44567875c79a / Work Public Subnet
☐	lab-rtb-public	rtb-026c49a9fe771df02	subnet-083a1f41f5545776f / lab-subnet-public1-us-eas.
☒	lab-rtb-private1-us-east-1a	rtb-05c1b443624f417ac	subnet-0e598722cde6f2b7c / lab-subnet-private1-us-ea.
☐	-	rtb-0ea8583de8b9600e7	-
☐	-	rtb-078b7823aa6b1c9ff	-
☐	-	rtb-04be131bd86459569	-

Go to the routes tab and notice the 0.0.0.0/0 route point to the NAT gateway this means all unresolved traffic will go to the internet.

Details **Routes** Subnet associations Edge associations Route propagation Tags

Routes (2)

Both ▼

[Edit routes](#)

Q Filter routes

< 1 > ⚙️

Destination ▼	Target ▼	Status ▼	Propagated ▼	Route Origin ▼
0.0.0.0/0	nat-0845f92b9fee13...	✓ Active	No	Create Route
10.0.0.0/16	local	✓ Active	No	Create Route Table

Now go to the Subnet associations tab and notice how it is associated with only itself, but now we want to associate itself with the second private subnet as well.

Choose edit Subnet Associations and select the lab-subnet-private2

Edit subnet associations

Change which subnets are associated with this route table.

Available subnets (2/4)

Q Filter subnet associations

< 1 > ⚙️

☐	Name ▼	Subnet ID ▼	IPv4 CIDR ▼	IPv6 CIDR ▼	Route table ID
☐	lab-subnet-public1-us-east-1a	subnet-083a1f41f5545776f	10.0.0.0/24	-	rtb-026c49a9fe771df02 / lab-rtb-pu
☒	lab-subnet-public2	subnet-0ae3355fc815a5879	10.0.2.0/24	-	Main (rtb-0ea8583de8b9600e7)
☐	lab-subnet-private2	subnet-0787b6fa5bc18e750	10.0.3.0/24	-	Main (rtb-0ea8583de8b9600e7)
☒	lab-subnet-private1-us-east-...	subnet-0e598722cde6f2b7c	10.0.1.0/24	-	rtb-05c1b443624f417ac / lab-rtb-pr

Selected subnets

[subnet-0e598722cde6f2b7c / lab-subnet-private1-us-east-1a](#) ×
 [subnet-0ae3355fc815a5879 / lab-subnet-public2](#) ×

Cancel

[Save associations](#)

Choose Save Associations and repeat this process for the public subnets

Edit subnet associations

Change which subnets are associated with this route table.

Available subnets (2/4)

Q Filter subnet associations

< 1 > ⚙️

☒	Name ▼	Subnet ID ▼	IPv4 CIDR ▼	IPv6 CIDR ▼	Route table ID
☒	lab-subnet-public1-us-east-1a	subnet-083a1f41f5545776f	10.0.0.0/24	-	rtb-026c49a9fe771df02 / lab-rtb-pu
☒	lab-subnet-public2	subnet-0ae3355fc815a5879	10.0.2.0/24	-	rtb-05c1b443624f417ac / lab-rtb-pr
☐	lab-subnet-private2	subnet-0787b6fa5bc18e750	10.0.3.0/24	-	Main (rtb-0ea8583de8b9600e7)
☐	lab-subnet-private1-us-east-...	subnet-0e598722cde6f2b7c	10.0.1.0/24	-	rtb-05c1b443624f417ac / lab-rtb-pr

Selected subnets

[subnet-083a1f41f5545776f / lab-subnet-public1-us-east-1a](#) ×
 [subnet-0ae3355fc815a5879 / lab-subnet-public2](#) ×

Cancel

[Save associations](#)

Next you will create a VPC security group which acts as a virtual firewall. When you launch an instance, you associate one or more security groups with the instance. You can add rules to each security group that allow traffic to or from its associated instances.

In the left navigation pane choose Security Groups

Choose Create Security Group and configure the following

Create security group [Info](#)

A security group acts as a virtual firewall for your instance to control inbound and outbound traffic. To create a new security group, complete the fields below.

Basic details

Security group name [Info](#)

Name cannot be edited after creation.

Description [Info](#)

VPC [Info](#)

In the Inbound Rules pane, choose Add Rule and configure the following

Inbound rules [Info](#)

Type Info	Protocol Info	Port range Info	Source Info	Description - optional Info
HTTP	TCP	80	An... 0/0	Permit web requests
Add rule Custom Anywhere-IPv4 ✓ Anywhere-IPv6 My IP				

Rules with source of 0.0.0.0/0 or ::/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.

Scroll down and choose Create Security Group

Now you will launch a Web server Instance

Use the console search bar and search EC2 and click on EC2

From the Launch Instance menu choose Launch instance

Name the instance: Web Server 1

Choose the default Amazon Linux 2023 AMI selected

Launch an instance [Info](#)

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

Name and tags [Info](#)

Name

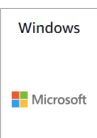
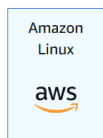
[Add additional tags](#)

Application and OS Images (Amazon Machine Image) [Info](#)

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose [Browse more AMIs](#).

Recents

Quick Start



[Browse more AMIs](#)
Including AMIs from
AWS, Marketplace and
the Community

Then in the Instance Type Choose t2.micro

▼ **Instance type** [Info](#) | [Get advice](#)

Instance type

t2.micro

Family: t2 1 vCPU 1 GiB Memory Current generation: true

On-Demand Windows base pricing: 0.0162 USD per Hour

On-Demand Ubuntu Pro base pricing: 0.0134 USD per Hour

On-Demand SUSE base pricing: 0.0116 USD per Hour

On-Demand RHEL base pricing: 0.026 USD per Hour

On-Demand Linux base pricing: 0.0116 USD per Hour

☐ All generations

[Compare instance types](#)

[Additional costs apply for AMIs with pre-installed software](#)

From the Key pair name menu choose vockey

▼ **Key pair (login)** [Info](#)

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - required

vockey



[Create new key pair](#)

Configure the Network Settings, next to the Network settings Choose Edit and configure:

Network: lab-vpc

Subnet: lab-subnet-public2 (not Private!)

Auto-assign public IP: Enable

Next Assign it the security group created earlier

Under Firewall (security groups), choose Select existing security groups.

For Common Security Groups, select Web Security Group.

▼ Network settings

Info

VPC - required

Info

vpc-09fbc6db0db097363 (lab-vpc)

10.0.0.0/16

↻

Subnet

Info

subnet-0ae3355fc815a5879

lab-subnet-public2

VPC: vpc-09fbc6db0db097363

Owner: 902178220333

Availability Zone: us-east-1b (use1-az1)

Zone type: Availability Zone

IP addresses available: 251

CIDR: 10.0.2.0/24

↻

Create new subnet

Auto-assign public IP

Info

Enable

▼

Additional charges apply

when outside of free tier allowance

Firewall (security groups)

Info

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☐ Create security group

☒ Select existing security group

Common security groups

Info

Select security groups

▼

Web Security Group sg-061a04b6a1252f6fd

×

VPC: vpc-09fbc6db0db097363

↻

Compare security group rules

Security groups that you add or remove here will be added to or removed from all your network interfaces.

► Advanced network configuration

In the configure storage setting keep everything to its default settings
 Configure a script to run on the instance when it launches:
 Expand the Advanced Details Panel and scroll to the bottom of the page and copy paste the following code:

```
#!/bin/bash
# Install Apache Web Server and PHP
dnf install -y httpd wget php mariadb105-server
# Download Lab files
wget https://aws-tc-largeobjects.s3.us-west-2.amazonaws.com/CUR-TF-100-ACCLFO-2/2-lab2-vpc/s3/lab-app.zip
unzip lab-app.zip -d /var/www/html/
# Turn on web server
chkconfig httpd on
service httpd start
```

User data - *optional* | [Info](#)

Upload a file with your user data or enter it in the field.

 [Choose file](#)

```
#!/bin/bash
# Install Apache Web Server and PHP
dnf install -y httpd wget php mariadb105-server
# Download Lab files
wget https://aws-tc-largeobjects.s3.us-west-2.amazonaws.com/CUR-TF-100-ACCLFO-2/2-lab2-vpc/s3/lab-app.zip
unzip lab-app.zip -d /var/www/html/
# Turn on web server
chkconfig httpd on
service httpd start
```

☐ User data has already been base64 encoded

Now Choose Launch Instance, then choose to view all instances and wait until Web Server 1 shows 2/2 checks passed, then select the web server and copy the public DNS value

i-040d35f1262d65e2f (Web Server 1)



[Details](#)

[Status and alarms](#)

[Monitoring](#)

[Security](#)

[Networking](#)

[Storage](#)

[Tags](#)

▼ Instance summary [Info](#)

Instance ID

 i-040d35f1262d65e2f

Public IPv4 address

 3.208.86.216 | [open address](#) 

Private IPv4 addresses

 10.0.2.220

IPv6 address

—

Instance state

 **Running**

Public DNS


ec2-3-208-86-216.compute-1.amazonaws.com
| [open address](#) 

Open the Link in a new tab. You should see a web page displaying the AWS logo and instance meta-data values.

aws

Load Test

RDS

Meta-Data	Value
InstanceId	i-040d35f1262d65e2f
Availability Zone	us-east-1b

Current CPU Load: 0%

The complete architecture deployed is:

aws

AWS Cloud

Region

Availability Zone A

lab-vpc: 10.0.0.0/16

lab-subnet-public1: 10.0.0.0/24

NAT gateway

lab-subnet-private1: 10.0.1.0/24

Availability Zone B

lab-subnet-public2: 10.0.2.0/24

Security group

Web Server 1

lab-subnet-private2: 10.0.3.0/24

Internet gateway

Public Route Table

Destination	Target
10.0.0.0/16	local
0.0.0.0/0	Internet gateway

Private Route Tables

Destination	Target
10.0.0.0/16	local
0.0.0.0/0	NAT gateway

Now Submit the work for a full score:

Total score	30/30
Task 1 - VPC created correctly	5/5
Task 2a - New subnets created correctly	5/5
Task 2b - Subnet route table association	5/5
Task 3 - Security group created correctly	5/5
Task 4a - EC2 instance created correctly	5/5
Task 4b - EC2 instance website accessible	5/5

Lab 3 – Introduction to Amazon EC2

In this lab you will be taught the basic overview of launching, resizing, managing and monitoring an Amazon EC2 Instance

Launch the Lab using the steps from Lab 1 instructions and make sure your region is set to N. Virginia (us-east-1) using the steps covered in Lab 1.

Now Navigate to EC2 using the search bar on the top left of the console. Double check between every step if you are in the correct region.

Choose the Launch Instance Menu and Choose Launch Instance.

Launch instance

To get started, launch an Amazon EC2 instance, which is a virtual server in the cloud.

Launch instance

▼

Migrate a server

Note: Your instances will launch in the United States (N. Virginia) Region

Name the Instance: Web Server

This name will be stored as a tag with a key of “name” and value of “Web Server”

In the quick start list choose Amazon Linux

Name and tags Info

Name

Web Server

Add additional tags

▼ Application and OS Images (Amazon Machine Image) Info

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose [Browse more AMIs](#).

Search our full catalog including 1000s of application and OS images

Recents

Quick Start

Amazon Linux

aws

macOS

Mac

Ubuntu

ubuntu

Windows

Microsoft

Red Hat

Red Hat

SUSE Linux

SUSE

Search

Browse more AMIs

Including AMIs from AWS, Marketplace and the Community

Amazon Machine Image (AMI)

Amazon Linux 2023 kernel-6.1 AMI
ami-0532be01f26a3de55 (64-bit (x86), uefi-preferred) / ami-0bb7267a511c0a8e8 (64-bit (Arm), uefi)
Virtualization: hvm ENA enabled: true Root device type: ebs

Description

Amazon Linux 2023 (kernel-6.1) is a modern, general purpose Linux-based OS that comes with 5 years of long term support. It is optimized for AWS and designed to provide a secure, stable and high-performance execution environment to develop and run your cloud applications.

Amazon Linux 2023 AMI 2023.10.20260120.4 x86_64 HVM kernel-6.1

Architecture	Boot mode	AMI ID	Publish Date	Username
64-bit ...	uefi-preferred	ami-0532be01f26a3de55	2026-01-22	ec2-user

Verified provider

In the instance type choose t2.micro

For the Key pair (login) name it “vockey”

▼ Instance type [Info](#) | [Get advice](#)

Instance type

t2.micro

Family: t2 1 vCPU 1 GiB Memory Current generation: true

On-Demand Windows base pricing: 0.0162 USD per Hour

On-Demand Ubuntu Pro base pricing: 0.0134 USD per Hour

On-Demand SUSE base pricing: 0.0116 USD per Hour

On-Demand RHEL base pricing: 0.026 USD per Hour

On-Demand Linux base pricing: 0.0116 USD per Hour

☐ All generations

[Compare instance types](#)

[Additional costs apply for AMIs with pre-installed software](#)

▼ Key pair (login) [Info](#)

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - *required*

vockey



[Create new key pair](#)

Go into the Network Setting by pressing Edit

Choose the pre-made VPC for you (Lab VPC)

Also choose the created subnet: PublicSubnet1

Under Firewall (security groups create a new security group, this will determine what the EC2 Instance can access and what can access the EC2 Instance.

For this Security group we will remove all inbound security groups to prevent all access as security groups only specify allowed connections and will deny anything not allowed.

Configure the following for the Security Group:

Name: Web Server security group

Description: Security group for my web server

Remove the default inbound rule

Keep the default storage settings. The default is 8GB of disk volume and is known as the root volume. These disks are network-attached virtual disks called the Elastic block store.

▼ Network settings [Info](#)

VPC - *required* [Info](#)

vpc-0df2f16647679f414 (Lab VPC)
10.0.0.0/16



Subnet [Info](#)

subnet-0aa8e40e3e850c8ed **PublicSubnet1**
VPC: vpc-0df2f16647679f414 Owner: 321033141866
Availability Zone: us-east-1a (use1-az2) Zone type: Availability Zone
IP addresses available: 1 CIDR: 10.0.1.0/28



[Create new subnet](#)

Auto-assign public IP [Info](#)

Enable

[Additional charges apply](#) when outside of [free tier allowance](#)

Firewall (security groups) [Info](#)

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☒ Create security group

☐ Select existing security group

Security group name - *required*

Web Server security group

This security group will be added to all network interfaces. The name can't be edited after the security group is created. Max length is 255 characters. Valid characters: a-z, A-Z, 0-9, spaces, and `._:/()#,@[]+=&;{}!$*`

Description - *required* [Info](#)

Security group for my web server

Inbound Security Group Rules

No security group rules are currently included in this template. Add a new rule to include it in the launch template.

[Add security group rule](#)

► **Advanced network configuration**

Towards the bottom, expand the Advanced details. Enable Termination protection, this will prevent accidental termination of an instance.

Termination protection [Info](#)

Enable



Scroll to the bottom now and copy paste the code below into the User Data box:

```
#!/bin/bash
dnf install -y httpd
systemctl enable httpd
systemctl start httpd
echo '<html><h1>Hello From Your Web Server!</h1></html>' > /var/www/html/index.html
```

This code will automatically run when the instance is launched to automate tasks. This script will do the following:

- Install an Apache web server (httpd)
- Configure the web server to automatically start on boot
- Run the Web server once it has finished installing
- Create a simple web page

User data - *optional* | [Info](#)

Upload a file with your user data or enter it in the field.

 Choose file

```
#!/bin/bash
dnf install -y httpd
systemctl enable httpd
systemctl start httpd
echo '<html><h1>Hello From Your Web Server!</h1></html>' >
/var/www/html/index.html
```

Now launch the instance by using the Launch instance button at the bottom of the console.

Now once the instance is created choose the View all Instances Button and select the Web Server you just created, it will display all the information about the instance you just created, the public IPv4 DNS is what you can use to connect to the Web Server.

Now Wait for the instance to pass all status checks and for the instance state to say “Running”

While the Instance is still selected navigate to the Status Checks tab and notice how the System reachability check and Instance reachability check passed.

i-0c596a9b3f02dcda5 (Web Server)



Details **Status and alarms** Monitoring Security Networking Storage Tags

Status checks [Info](#)

Actions ▾

Status checks detect problems that may impair i-0c596a9b3f02dcda5 (Web Server) from running your applications.

System status checks

✔ System reachability check passed

Instance status checks

✔ Instance reachability check passed

► Metrics

▼ Alarms

ℹ Recently launched instances can take up to 5 minutes to display associated alarms.

🔍 Find alarms by name

< 1 > ⚙

Name	State	Description	Metric name	State reason
------	-------	-------------	-------------	--------------

Instance has no associated alarms

Now Choose the Monitoring tab, this displays all the CloudWatch Metrics for instance, you can choose the 3-dot icon on any graph to enlarge the data shown in the given metric. You will receive a No data available message if your instance was just launched.

i-0c596a9b3f02dcda5 (Web Server)



Details **Status and alarms** **Monitoring** Security Networking Storage Tags

☒ Include metrics in the CWAgent namespace ⓘ
[Learn more](#) ↗

Configure CloudWatch agent

Manage detailed monitoring

☐ Alarm recommendations ⓘ

3h 1d 1w 1h 📅

UTC timezone ▼



🔍 Explore related



CPU utilization (%) ⓘ ⋮

No data available.

Feb 4, 19:00 Feb 4, 19:30

Network in (bytes) ⓘ ⋮

No data available.

Feb 4, 19:00 Feb 4, 19:30

Network out (bytes) ⓘ ⋮

No data available.

Feb 4, 19:00 Feb 4, 19:30

Network packets in (count) ⓘ ⋮

No data available.

Feb 4, 19:00 Feb 4, 19:30

Network packets out (count) ⓘ ⋮

No data available.

Feb 4, 19:00 Feb 4, 19:30

Metadata no token (count) ⓘ ⋮

No data available.

Feb 4, 19:00 Feb 4, 19:30

CPU credit usage (count) ⓘ ⋮

No data available.

Feb 4, 19:00 Feb 4, 19:30

CPU credit balance (count) ⓘ ⋮

No data available.

Feb 4, 19:00 Feb 4, 19:30

In the Actions menu towards the top left of the console menu choose Monitor and Troubleshoot > Get System Log. This displays the system log and is important when troubleshooting Instance errors and malfunctions. When selecting this you can

also see the HTTP package was installed in the instance from the code imputed into the user data.

Instances (1/2) Info Last updated 1 minute ago [Connect](#) [Instance state](#) [Actions](#) [Launch instances](#)

Find Instance by attribute or tag (case-sensitive)

	Name	Instance ID	Instance state	Instance type
☐	Bastion Host	i-0c7851e4861b5375a	Running	t2.micro
☒	Web Server	i-0c596a9b3f02dcda5	Running	t2.micro

i-0c596a9b3f02dcda5 (Web Server)

Details | Status and alarms | **Monitoring**

☒ Include metrics in the CWAgent namespace [Learn more](#)

☐ Alarm recommendations

CPU utilization (%) | Network in (bytes) | Network out (bytes) | Network packets in (count)

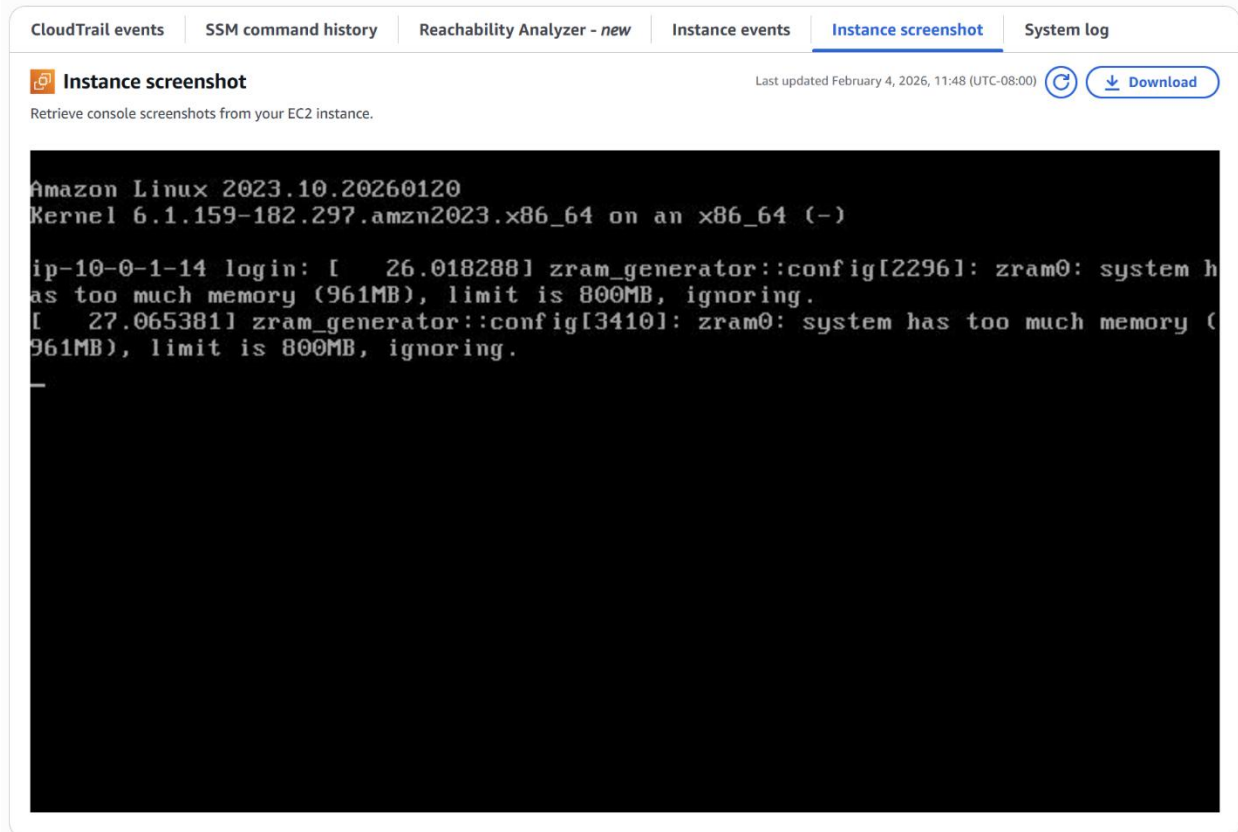
System log [Copy log](#) [Download](#)

Review system log for instance i-04a8042ed8eed9a93 as of Thu Mar 23 2023 09:36:29 GMT-0700 (Pacific Daylight Time)

```
[ 56.398498] cloud-init[2068]: apr-util-1.6.3-1.amzn2023.0.1.x86_64
[ 56.408260] cloud-init[2068]: apr-util-openssl-1.6.3-1.amzn2023.0.1.x86_64
[ 56.429206] cloud-init[2068]: generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch
[ 56.438449] cloud-init[2068]: httpd-2.4.55-1.amzn2023.x86_64
[ 56.451350] cloud-init[2068]: httpd-core-2.4.55-1.amzn2023.x86_64
[ 56.469399] cloud-init[2068]: httpd-filesystem-2.4.55-1.amzn2023.noarch
[ 56.477389] cloud-init[2068]: httpd-tools-2.4.55-1.amzn2023.x86_64
[ 56.489079] cloud-init[2068]: libbrotli-1.0.9-4.amzn2023.0.2.x86_64
[ 56.494014] cloud-init[2068]: mailcap-2.1.49-3.amzn2023.0.3.noarch
[ 56.502033] cloud-init[2068]: mod_http2-2.0.11-2.amzn2023.x86_64
[ 56.510985] cloud-init[2068]: mod_lua-2.4.55-1.amzn2023.x86_64
[ 56.529296] cloud-init[2068]: Complete!
[ 56.540237] cloud-init[2068]: Created symlink /etc/systemd/system/multi-user.target.wants/httpd.service â /usr/lib/systemd/system/httpd.service.
[ 56.569490] systemd-sysv-generator[3273]: SysV service '/etc/rc.d/init.d/cfn-hup' lacks a native systemd unit file. Automatically generating a unit file.
ci-info: +-----+Authorized keys from /home/ec2-user/.ssh/authorized_keys for user ec2-user+-----+
ci-info: | Keytype |                               Fingerprint (sha256)                               | Options | Comment |
ci-info: +-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
ci-info: | ssh-rsa | c0:8a:46:b6:e4:be:f4:ec:79:81:da:f9:c4:14:a8:db:c5:cb:8c:ae:18:50:3e:70:1b:fd:59:91:e1:78:77:93 | - | vockey |
ci-info: +-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+

```

Choose Cancel and make sure the Web Serve instance is still selected. Then in the Actions menu go to Monitor and troubleshoot > Get instance screenshot. This shows you what your instance looks like currently.



Make sure the Web Server is still selected and choose the Details tab to get the Public IPv4 address copied to your clipboard.

If you try pasting this URL you will notice it does not work, this is because of the security group we configured to deny access earlier. (Note that we did not configure any inbound rules for the security group which is why traffic is being denied as unless explicitly permitted, traffic is implicitly denied).

Keep the tab with the EC2 instance URL open and go back to the tab with the console. In the left navigation pane choose Security Groups.

Select the security group created during the instance Creation: Web security group.

Go to the inbound rules tab and select Edit Inbound rules, and select add rule and configure the rule for the following:

- Type: HTTP
- Source: Anywhere IPv4
- Choose: Save Rules

Go back to the tab open to the web server and refresh, you should see a message: *Hello From Your Web Server!*

Hello From Your Web Server!

As your needs change, you might find that your instance is over-utilized (too small) or under-utilized (too large). If so, you can change the instance type.

Before you can resize an instance, you must stop it.

In the EC2 Console go to the Instance State Menu and select Stop Instance while the Web Server Instance is selected. Wait for the Instance to display stopped.

Keep the Web server Instance selected and go to the actions Menu and select Instance Settings > Change Instance Type, and configure the instance type to be t2.small. and choose Apply.

Instances (1/2) Info Last updated 12 minutes ago

Find Instance by attribute or tag (case-sensitive)

	Name	Instance ID	Instance state	Health status check	Alarm status
☐	Bastion Host	i-0c7851e4861b5375a	Running	2/2 checks passed	View alarms +
☒	Web Server	i-0c596a9b3f02dcda5	Running	2/2 checks passed	View alarms +

Stop instance
Start instance
Reboot instance
Hibernate instance
Terminate (delete) instance

Successfully initiated stopping of i-0c596a9b3f02dcda5

Instances (1/2) Info Last updated less than a minute ago

Find Instance by attribute or tag (case-sensitive)

	Name	Instance ID
☐	Bastion Host	i-0c7851e4861b5375a
☒	Web Server	i-0c596a9b3f02dcda5

Instance diagnostics
Instance settings
Networking
Security
Image and templates
Storage
Monitor and troubleshoot
Attach to Auto Scaling Group
Change termination protection
Change stop protection
Change shutdown behavior
Change instance migration on reboot
Change auto-recovery behavior
Change instance type

Instance ID
i-0c596a9b3f02dcda5 (Web Server)

Current instance type
t2.small

New instance type
t2.small

Select the Web Server instance, then in the Actions menu, select Instance settings > Change stop protection. Select Enable and then Save the change.

Instance type changed successfully

Instances (1/2) Info Last updated 2 minutes ago [Refresh](#) [Connect](#) [Instance state](#) [Actions](#) [Launch instances](#)

Find Instance by attribute or tag (case-sensitive)

Name	Instance ID
Bastion Host	i-0c7851e4861b5375a
Web Server	i-0c596a9b3f02dcda5

i-0c596a9b3f02dcda5 (Web Server)

Attach to Auto Scaling Group
Change termination protection
Change stop protection
Change shutdown behavior
Change instance migration on reboot
Change auto-recovery behavior
Change instance type

Instance diagnostics
Instance settings
Networking
Security
Image and templates
Storage
Monitor and troubleshoot

Now if you need more storage in the EC2 instance you can resize the EBS or Elastic Block Store Volume.

With the Web Server Instance still selected go the storage tab and select the Volume ID listed.

In the actions menu select Modify Volume.

Volumes (1/1) Info Last updated less than a minute ago [Refresh](#) [Recycle Bin](#) [Actions](#) [Create volume](#)

Saved filter sets Choose filter set Search

Volume ID = vol-02c0e59244c7e37b9 Clear filters

Name	Volume ID	Type	Size
	vol-02c0e59244c7e37b9	gp3	8 GiB

Modify volume
Create snapshot
Create snapshot lifecycle policy
Delete volume
Attach volume
Detach volume
Force detach volume
Manage auto-enabled I/O
Copy volume - new
Manage tags
Resilience testing
Previously fault injection

Change the size of the disk from 8 > 10GB

Modify volume Info

Modify the type, size, and performance of an EBS volume.

Volume details

Volume ID
vol-02c0e59244c7e37b9

Volume type Info
General Purpose SSD (gp3)

Size (GiB) Info
10
Min: 1 GiB, Max: 65536 GiB.

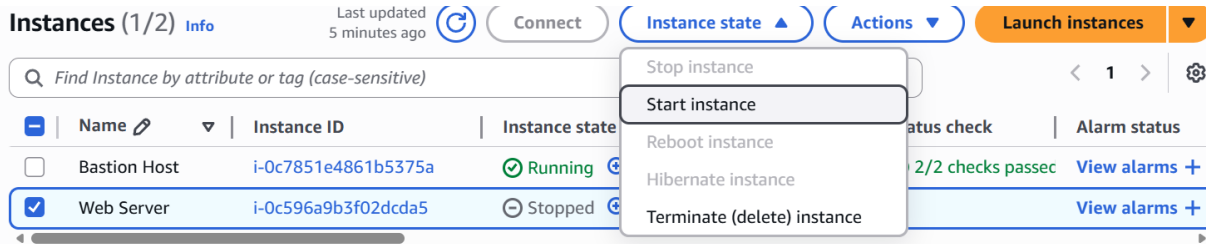
IOPS Info
3000
Min: 3000 IOPS, Max: 80000 IOPS.

Throughput (MiB/s) Info
125
Min: 125 MiB, Max: 2000 MiB. Baseline: 125 MiB/s.

[Cancel](#) [Modify](#)

Choose Modify, and confirm your choice.

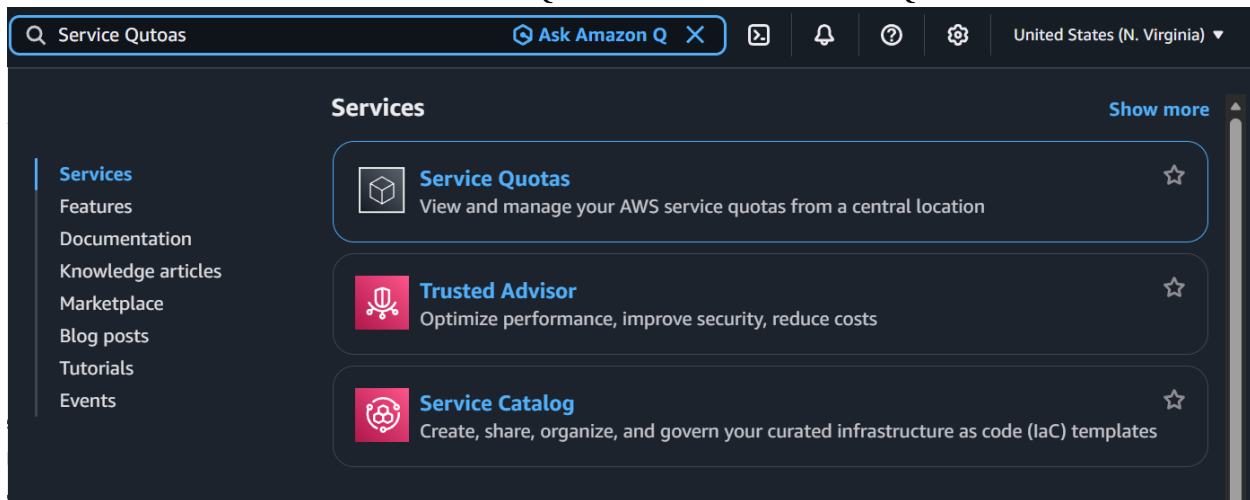
Now Select the Web server instance and start it



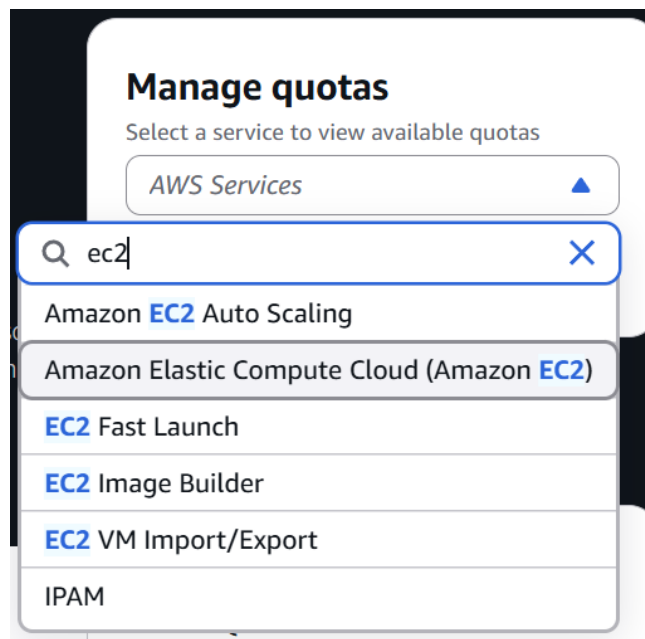
You have now resized the instance type and volume of the EC2 instance

When you create an AWS account there are per region limitations on how many EC2 instances you can own. In this next part we will explore those limitations.

In the console search Service Quotas and click Service Quotas



In the Manage Quotas box search EC2 and choose Amazon Elastic Compute Cloud.



Now search running on-demand in the service quotas search box and just notice all the account limitations that the AWS account has for on-demand EC2 Instances like the Web Server EC2 instance you created.

Service quotas info Request increase at account level

View your applied quota values, default quota values, and request quota increases for quotas. [Learn more](#)

Q running on-demand X 10 matches < 1 > ⚙

	Quota name ▲	Applied account-level quota value ▼	AWS default quota value ▼	Utilization ▼	Adjustability ▼
☐	Running On-Demand DL instances	96	0	0	Account level
☐	Running On-Demand F instances	64	0	0	Account level
☐	Running On-Demand G and VT instances	0	0	0	Account level
☐	Running On-Demand High Memory instances	0	0	0	Account level
☐	Running On-Demand HPC instances	192	0	0	Account level
☐	Running On-Demand Inf instances	8	0	0	Account level
☐	Running On-Demand P instances	0	0	0	Account level
☐	Running On-Demand Standard (A, C, D, H, I, M, R, T, Z) instances	256	5	1	Account level
☐	Running On-Demand Trn instances	8	0	0	Account level
☐	Running On-Demand X instances	0	0	0	Account level

Now let's test the stop protection on the EC2 instance we created. Navigate back to the EC2 Console and try stopping the EC2 instance.

Instances (1/2) Info Refresh Connect Instance state ▲ Actions ▼ Launch instances ▼

Q Find Instance by attribute or tag (case-sensitive)

	Name	Instance ID	Instance state	Health status check	Alarm status
☐	Bastion Host	i-0c7851e4861b5375a	Running	2/2 checks passed	View alarms +
☒	Web Server	i-0c596a9b3f02dcda5	Running	Initializing	View alarms +

Stop instance

Start instance

Reboot instance

Hibernate instance

Terminate (delete) instance

Notice that you will get an error message saying you failed to stop the instance

Instances (1/2) Info Refresh Connect Instance state ▲ Actions ▼ Launch instances ▼

Q Find Instance by attribute or tag (case-sensitive)

	Name	Instance ID	Instance state	Health status check	Alarm status
☐	Bastion Host	i-0c7851e4861b5375a	Running	2/2 checks passed	View alarms +
☒	Web Server	i-0c596a9b3f02dcda5	Running	Initializing	View alarms +

Stop instance

Start instance

Reboot instance

Hibernate instance

Terminate (delete) instance

No remove the Stop protection the same way you enabled it but this time unchecking the enable checkbox.

Go to Instance settings > Change Stop Protection.

Change stop protection

Enable stop protection to prevent your instance from being accidentally stopped.
[Learn more](#)

Instance ID
 i-0c596a9b3f02dcda5 (Web Server)

Stop protection
☐ Enable

[Cancel](#) [Save](#)

Now you've completed your lab, submit to the lab to receive a full score.

Total score	25/25
Task 1 - EC2 instance created correctly	5/5
Task 2 - get system log requested	5/5
Task 3 - security group updated	5/5
Task 4 - EC2 instance updated	5/5
Task 6 - Instance stopped on second try	5/5

Conclusion

This lab has helped develop the foundation of how to develop cloud solutions using AWS services like IAM, EC2, and VPCs. This additionally helps to build fundamental concepts within cloud computing like on-demand computing access and cloud security using the principle of least privilege. This lab additionally helps to get an understanding of how to configure and create cloud solutions using services other than AWS like Microsoft Azure. All in all, these 3 labs served as a great introduction to how to configure and create cloud solutions.